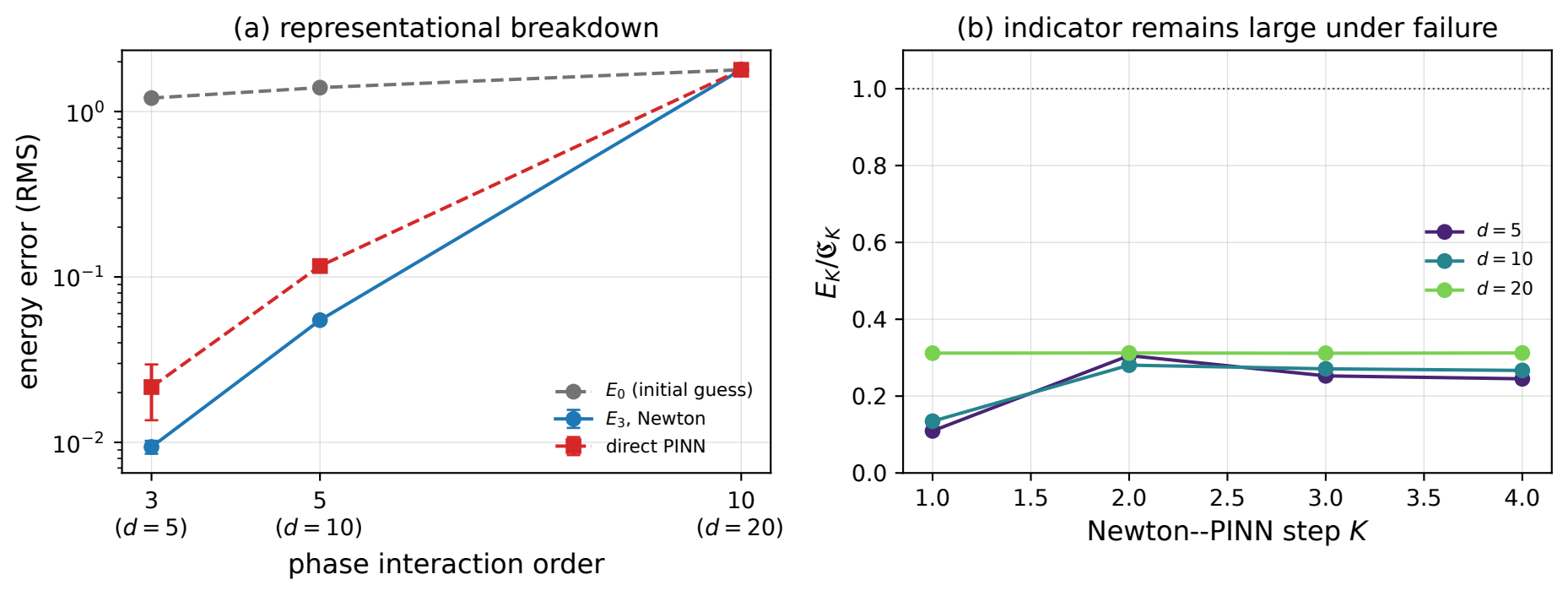
(a) representational breakdown
energy error (RMS)
10^0
10^{-1}
10^{-2}
E_0 (initial guess)
E_3, Newton
direct PINN
3
($d = 5$)
5
($d = 10$)
10
($d = 20$)
phase interaction order
(b) indicator remains large under failure
$E_K/\mathfrak{S}_K$
1.0
0.8
0.6
0.4
0.2
0.0
$d = 5$
$d = 10$
$d = 20$
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Newton--PINN step K